



Hospital Management System (HMS) - Database Schema

This document outlines the relational database structure required to support the Life Care Diagnostic Center's workflows, role management, billing, and patient tracking.

Entity-Relationship Overview

- **Users & Roles:** Controls access (Role Management).
- **Patients & Admissions:** Tracks demographics, Current Status , and hospital stays.
- **Beds:** Manages Ward, cabin, empty cabin (list) .
- **Clinical:** Manages Appointments and Operation date + preparation .
- **Pharmacy & Billing:** Connects Medicine -> Pharmacy and consolidates master Bills with Discounts for Admin Reports .

1. Access Control & Staff (Role Management)

roles

Defines the access levels (Admin, Doctor, Receptionist, Pharmacist, OT Manager, Cashier).

- `id` (PK) - INT
- `role_name` - VARCHAR (e.g., 'Doctor', 'Admin')
- `permissions` - JSON (Specific feature toggles)

users

The hospital staff members logging into the system.

- `id` (PK) - INT
- `role_id` (FK) - INT (Links to `roles.id`)
- `username` - VARCHAR
- `password_hash` - VARCHAR
- `full_name` - VARCHAR
- `department` - VARCHAR (e.g., 'Cardiology', 'Pharmacy')

2. Patient Flow & Facility Management

patients

Core patient demographic data.

- `id` (PK) - INT
- `uhid` - VARCHAR (Unique Hospital Identification Number)
- `full_name` - VARCHAR
- `contact_info` - VARCHAR
- `current_status` - VARCHAR (e.g., 'Waiting', 'In Consultation', 'Admitted', 'Cleared for Discharge')

beds

Tracks the **Ward, cabin, empty cabin (list)**.

- `id` (PK) - INT
- `bed_number` - VARCHAR (e.g., 'Cabin-101', 'Ward-A-Bed-05')
- `type` - VARCHAR ('Cabin', 'General Ward', 'ICU')
- `status` - VARCHAR ('Available', 'Occupied', 'Maintenance') - *Filtering by 'Available' generates the Empty Cabin list.*
- `daily_rate` - DECIMAL

admissions

Handles the **Admission** process.

- `id` (PK) - INT
- `patient_id` (FK) - INT
- `bed_id` (FK) - INT (Links to `beds.id`)
- `admitting_doctor_id` (FK) - INT (Links to `users.id`)
- `admission_date` - DATETIME
- `discharge_date` - DATETIME (Nullable)

3. Clinical & Operations

appointments

Handles the **Appointment** system for OPD.

- `id` (PK) - INT
- `patient_id` (FK) - INT
- `doctor_id` (FK) - INT
- `appointment_datetime` - DATETIME
- `status` - VARCHAR ('Scheduled', 'Completed', 'Cancelled')

operations

Handles **Operation date + preparation**.

- `id` (PK) - INT
- `patient_id` (FK) - INT
- `surgeon_id` (FK) - INT (Links to `users.id`)
- `ot_room` - VARCHAR
- `operation_datetime` - DATETIME
- `preparation_checklist` - TEXT (e.g., 'Fasting 12h, Blood type O- arranged')
- `status` - VARCHAR ('Scheduled', 'In Surgery', 'Recovery', 'Completed')

4. Pharmacy & Billing

prescriptions (Medicine -> Pharmacy)

Digital prescriptions sent from Doctor to Pharmacist.

- `id` (PK) - INT

- patient_id (FK) - INT
- doctor_id (FK) - INT
- notes - TEXT
- status - VARCHAR ('Pending Pharmacy', 'Dispensed')

prescription_items

Individual medicines within a prescription.

- id (PK) - INT
- prescription_id (FK) - INT
- medicine_name - VARCHAR
- quantity - INT
- unit_price - DECIMAL

bills (Master Bill)

Handles **Bill + Discount** and generates data for **Admin Report (range) + profit**.

- id (PK) - INT
- patient_id (FK) - INT
- admission_id (FK) - INT (Nullable, if OPD only)
- subtotal_amount - DECIMAL
- discount_amount - DECIMAL
- discount_approved_by (FK) - INT (Links to users.id - required for large discounts)
- net_total - DECIMAL (Subtotal - Discount)
- status - VARCHAR ('Unpaid', 'Partially Paid', 'Paid')
- created_at - DATETIME (Used to filter Admin Reports by date range)

bill_items (Sub-bills)

Feeds into the master bill from different departments.

- id (PK) - INT
- bill_id (FK) - INT
- department - VARCHAR ('Pharmacy', 'Consultation', 'Cabin Rent', 'Operation')
- description - VARCHAR (e.g., 'Paracetamol 500mg', 'Cabin-101 (3 days)')
- amount - DECIMAL